

A Deep State Space Model for Rainfall-Runoff Simulations

Yihan Wang¹, Lujun Zhang¹, Annan Yu², N. Benjamin Erichson³, Tiantian Yang^{1*}

¹ School of Civil Engineering and Environmental Science, University of Oklahoma, Norman, OK, United States

² Center for Applied Mathematics, Cornell University, Ithaca, NY, United States

³ Lawrence Berkeley National Laboratory, Berkeley, CA, United States

Contents of this file

Text S1. S4D-FT hyperparameter sensitivity analysis and tuning guidance.

Figure S1. Sensitivity of S4D-FT NSE performance to variations in individual hyperparameters.

Figure S2. Sensitivity of S4D-FT KGE performance to variations in individual hyperparameters.

Figure S3. Locations of four example watersheds and their hydrographs in a single water year.

Figure S4. Relative-error distributions on the top 2% of observed streamflow for S4D-FT and LSTM.

Table S1. Hydrometeorological variables and static catchment attributes for DL training.

Table S2. LSTM training hyperparameters.

Table S3. SSMs (basic S4D and S4D-FT) training hyperparameters.

Table S4. Description of employed watershed characteristics.

Table S5. Percentage of watersheds with improved (first row) and decreased (second row) performance of S4D-FT.

Text S1. S4D-FT hyperparameter sensitivity analysis and tuning guidance

In this analysis, for each hyperparameter, we retrain the S4D-FT model while holding all others at their selected optimal values for a single random seed. The tested ranges for each hyperparameter, along with their descriptions and the final employed values, are presented in [Table S3](#). The NSE and KGE results of the hyperparameter sensitivity analysis are illustrated in [Figures S1 and S2](#), respectively. In [Figures S1 and S2](#), the gray bars represent the reference performance obtained from the final selected hyperparameters, whereas the blue bars represent the performance of the tested hyperparameters.

Across the tested ranges, NSE is more robust to hyperparameter changes than KGE, implying that seemingly small configuration shifts can affect KGE more noticeably than NSE. In addition, based on the relative sensitivities observed in [Figures S1 and S2](#), practical guidance for configuring S4D-FT efficiently is provided below:

1. Start with the structural hyperparameters of **d_model** and **n_layer** (analogous to hidden size and depth in LSTM) to determine the model capacity first, followed by **cfi** and **dropout**.
2. Stabilize the discretization-related parameters. Adjust **min_dt** and **max_dt**. Avoid extremes (too small **min_dt** or too large **max_dt**) that degraded both metrics.
3. Refine with optimization hyperparameters by co-tuning **batch_size**, **lr**, **weight decay** (and **wd**).

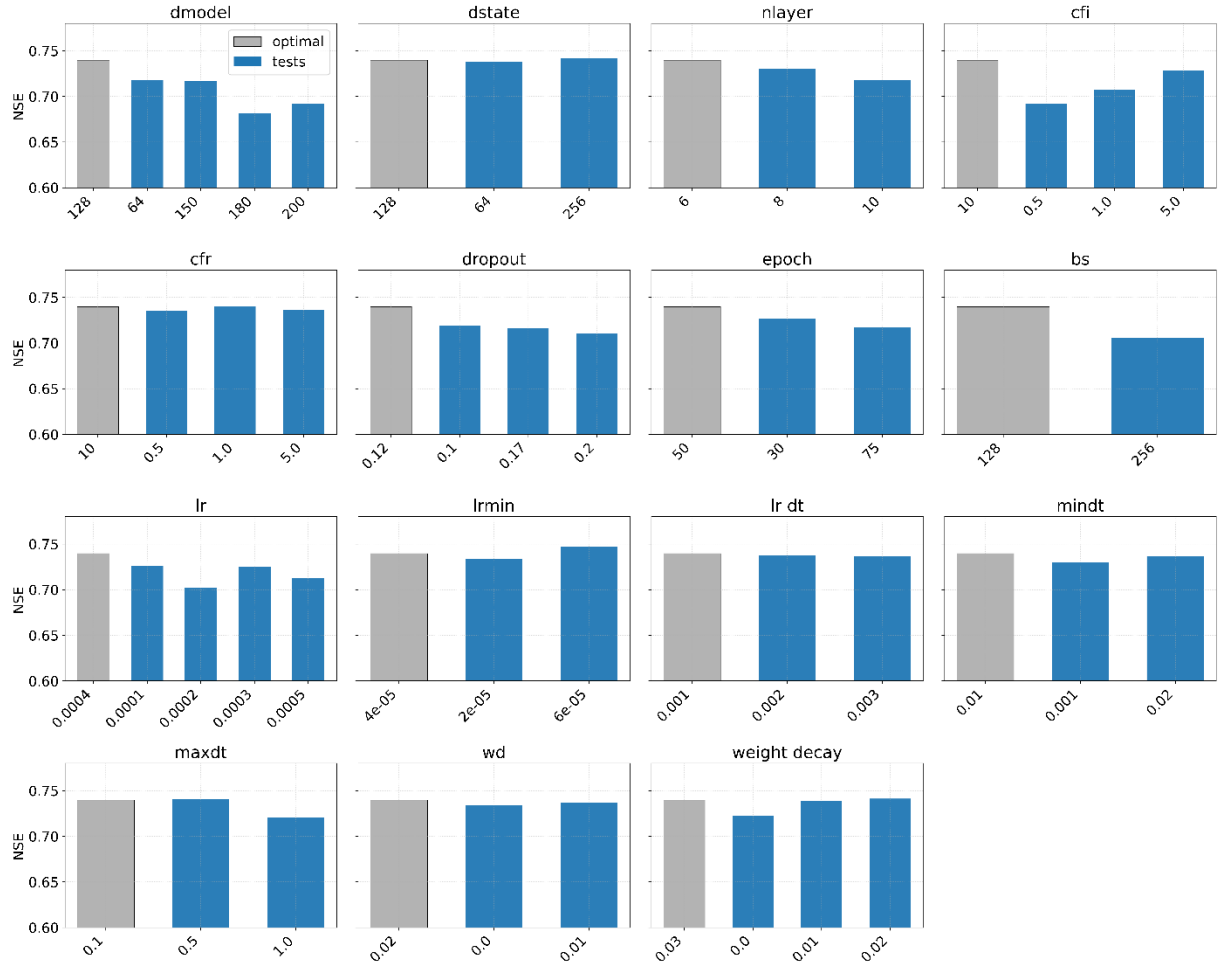


Figure S1. Sensitivity of S4D-FT NSE performance to variations in individual hyperparameters. The reference (optimal) performance is represented in gray bars, and the tests performance is represented in blue bars.

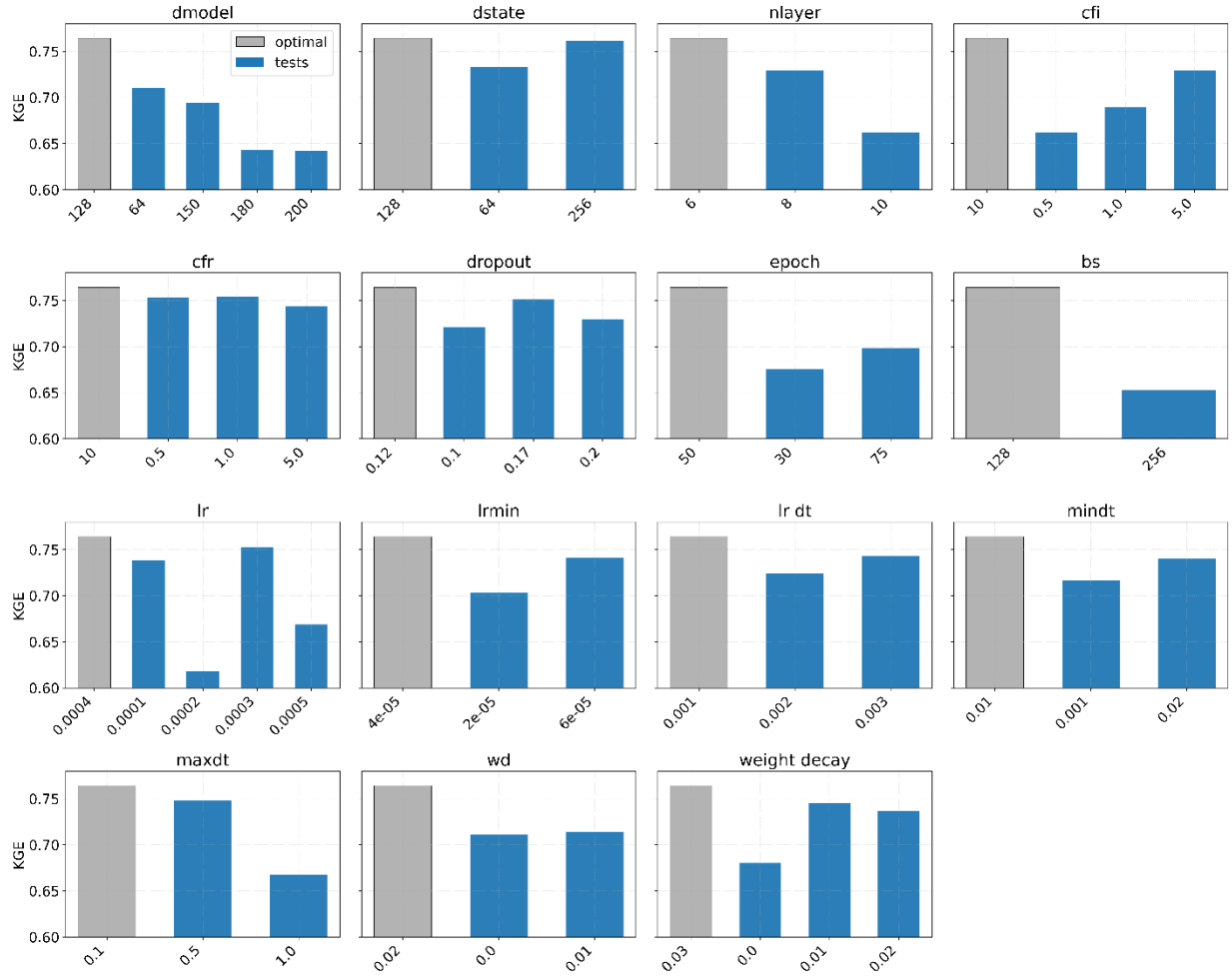


Figure S2. Sensitivity of S4D-FT KGE performance to variations in individual hyperparameters. The reference (optimal) performance is represented in gray bars, and the tests performance is represented in blue bars.

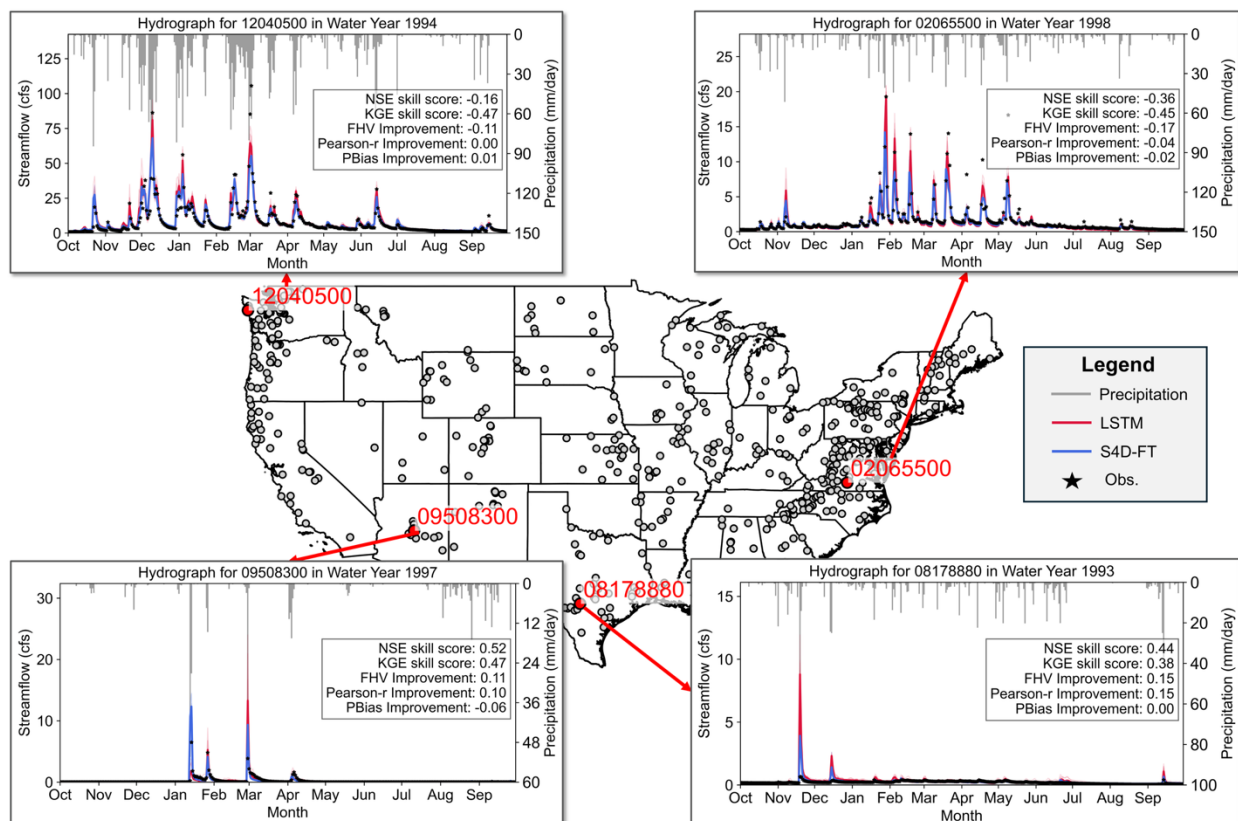


Figure S3. Locations of four example watersheds (center map) and their hydrographs in a single water year. The top row shows watersheds where S4D-FT underperforms LSTM (USGS 12040500 and 02065500); the bottom row shows watersheds where S4D-FT outperforms LSTM (USGS 09508300 and 08178880). Each panel shows simulated 8-member ensemble hydrographs for LSTM (red) and S4D-FT (blue), along with observed streamflow (black stars). Thin and thick lines indicate individual ensemble members and ensemble medians, respectively. Precipitation is shown on the secondary y-axis (gray bars). The NSE and KGE skill scores, along with improvements in FHV, Pearson-r, and PBias calculated from the testing results are displayed for each watershed.

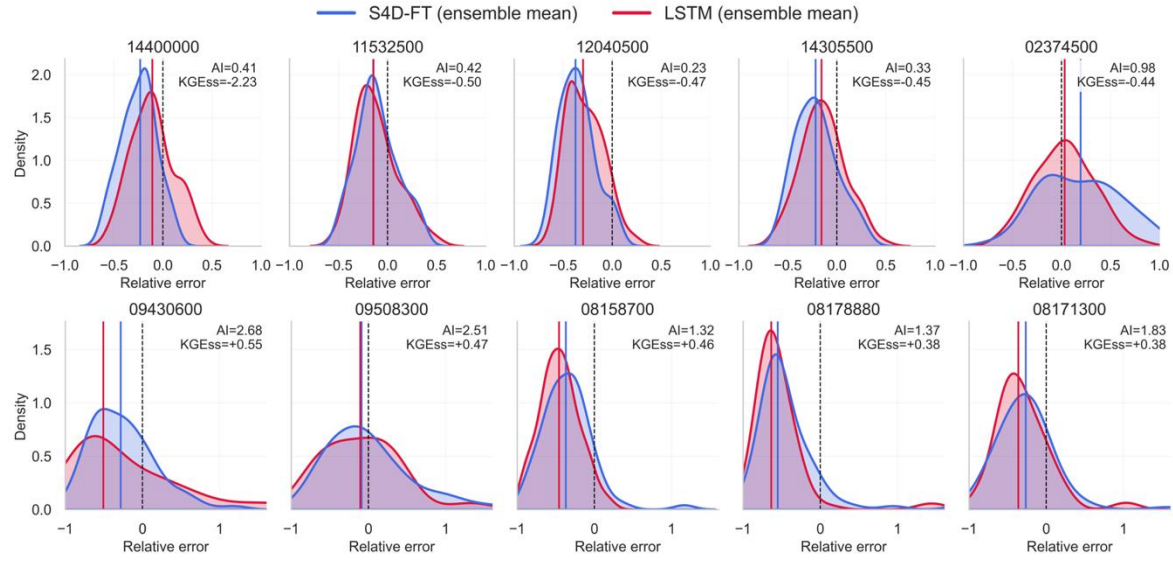


Figure S4. Relative-error distributions on the top 2% of observed streamflow for S4D-FT (blue) and LSTM (red) in representative watersheds. Top row: humid watersheds (Aridity index, AI < 1) where S4D-FT underperforms LSTM with the most negative KGE skill scores (KGEss). Bottom row: arid watersheds (AI > 1) where S4D-FT outperforms LSTM with the most positive KGEss. Colored vertical lines mark medians for each model. The black dashed line denotes zero bias. Station IDs, AI, and KGE skill (KGEss) are annotated in each panel.

Table S1. Hydrometeorological variables and static catchment attributes for DL training. Variable descriptions are adopted from [Kratzert et al. \(2019a\)](#).

Hydrometeorological Variables	1	Maximum air temp	2 m daily maximum air temperature (°C)
	2	Minimum air temp	2 m daily minimum air temperature (°C)
	3	Precipitation	Average daily precipitation (mm/day)
	4	Radiation	Surface-incident solar radiation (W/m ²)
	5	Vapor pressure	Near-surface daily average (P_a)
Static Catchment Attributes	6	Precipitation mean	Mean daily precipitation.
	7	PET mean	Mean daily potential evapotranspiration
	8	Aridity index	Ratio of Mean PET to Mean Precipitation
	9	Precip seasonality	Estimated by representing annual precipitation and temperature as sin waves. Positive (negative) values indicate precipitation peaks during the summer (winter). Values of ~0 indicate uniform precipitation throughout the year.
	10	Snow fraction	Fraction of precipitation falling on days with temp < 0 °C.
	11	High precipitation frequency	Frequency of days with $\leq 5 \times$ mean daily precipitation
	12	High precip duration	Average duration of high precipitation events (number of consecutive days with $\leq 5 \times$ mean daily precipitation).
	13	Low precip frequency	Frequency of dry days (< 1 mm/day).
	14	Low precip duration	Average duration of dry periods (number of consecutive days with precipitation < 1 mm/day).
	15	Elevation	Catchment mean elevation.
	16	Slope	Catchment mean slope.
	17	Area	Catchment area.
	18	Forest fraction	Fraction of catchment covered by forest.
	19	LAI max	Maximum monthly mean of leaf area index.
	20	LAI difference	Difference between the max. and min. mean of the leaf area index.
	21	GVF max	Maximum monthly mean of green vegetation fraction.
	22	GVF difference	Difference between the maximum and minimum monthly mean of the green vegetation fraction.
	23	Soil depth (Pelletier)	Depth to bedrock (maximum 50 m).
	24	Soil depth (STATSGO)	Soil depth (maximum 1.5 m).
	25	Soil Porosity	Volumetric porosity.
	26	Soil conductivity	Saturated hydraulic conductivity.
	27	Max water content	Maximum water content of the soil.

	28	Sand fraction	Fraction of sand in the soil.
	29	Silt fraction	Fraction of silt in the soil.
	30	Clay fraction	Fraction of clay in the soil.
	31	Carbonate rocks fraction	Fraction of the catchment area characterized as “carbonate sedimentary rocks”.
	32	Geological permeability	Surface permeability (log10).

Table S2. LSTM training hyperparameters.

Parameter	Value
hidden_size	256
batch_size	256
initial_forget_gate_bias	3
learning_rate	1e-3 for 1–10 epochs, 5e-4 for 11–25 epochs, and 1e-4 for 26–30 epochs
dropout	0.4
epochs	30

Table S3. SSMs (basic S4D and S4D-FT) training hyperparameters (employed values and tested values for sensitivity test) and description.

ID	parameter	Employed value	Description	Tested values
1	d_model	128	Number of channels per S4D layer.	64, 150, 180, 200
2	d_state	128	Number of latent SSM states per channel.	64, 256
3	n_layer	6	Number of stacked S4D layers.	8, 10
4	cfi (S4D-FT only)	10	Imaginary part scale of the state matrix $\mathbf{A}$	0.5, 1, 5
5	cfr (S4D-FT only)	10	Real part scale of the state matrix $\mathbf{A}$	0.5, 1, 5
6	ssm_dropout	0.12	Dropout rate applied within S4D layers to mitigate overfitting.	0.1, 0.17, 0.2
7	min_dt	0.01	Minimum discretization step-size in time discretization scheme.	0.001, 0.02
8	max_dt	0.1	Maximum discretization step-size in time discretization scheme.	0.5, 1
9	epochs	50	Number of training epochs.	30, 75
10	batch_size	128	Number of samples per training batch.	256
11	lr	4e-4	Peak global learning rate applied to non-SSM parameters.	1e-4, 2e-4, 3e-4, 5e-4
12	lr_min	4e-5	Learning rate used for S4D kernel parameters (A, B, C, Δt); the effective S4D lr is $\min(\text{lr}, \text{lr min})$	2e-5, 6e-5
13	lr_dt	0.001	Learning rate applied exclusively to the time-step parameter Δt within each S4D kernel.	0.002, 0.003
14	weight_decay	0.03	Global L2 regularization factor applied to all standard parameters except for a subset of S4D kernel weights, which are regularized separately.	0, 0.01, 0.02
15	wd	0.02	An additional L2 regularization specifically applied to the core state-space kernel parameters within each S4D (LTI) layer.	0, 0.01

Table S4. Description of employed watershed characteristics. This table is partially adopted from [Addor et al. \(2017\)](#).

Category	Variable	Description	Unit
Hydrologic signatures	q_mean	mean daily discharge	mm/day
	q5	5% flow quantile (low flow)	mm /day
	q95	95% flow quantile (high flow)	mm /day
	slope_fdc	slope of the flow duration curve (between the logtransformed 33rd and 66th streamflow percentiles)	-
	high_q_freq	frequency of high-flow days (> 9 times the median daily flow)	days/year
	high_q_dur	average duration of high-flow events (number of consecutive days > 9 times the median daily flow)	days
	low_q_freq	frequency of low-flow days (< 0.2 times the mean daily flow)	days/year
	low_q_dur	average duration of low-flow events (number of consecutive days < 0.2 times the mean daily flow)	days
	zero_q_freq	frequency of days with Q = 0	%
Climate	p_mean	mean daily precipitation	mm/day
	frac_snow	fraction of precipitation falling as snow (i.e., on days colder than 0 °C)	-
	aridity	ratio of mean PET to mean precipitation	-
	high_prec_dur	average duration of high precipitation events (number of consecutive days ≥ 5 times mean daily precipitation)	days
	low_prec_freq	frequency of dry days (< 1 mm/day)	days/year
	low_prec_dur	average duration of dry periods (number of consecutive days < 1 mm/day)	days
Soil	sand_frac	sand fraction (of the soil material smaller than 2mm)	%
	water_frac	fraction of the top 1.5m marked as water	%
	organic_frac	fraction of soil_depth_statsgo marked as organic material	%
Vegetation	lai_max	maximum monthly mean of the leaf area index (based on 12 monthly means)	-
	lai_diff	difference between the maximum and minimum monthly mean of the leaf area index (based on 12 monthly means)	-
	root_depth_50	root depth (percentiles 50 % extracted from a root depth distribution based on IGBP land cover)	m

Table S5. Percentage of watersheds with improved (first row) and decreased (second row) performance of S4D-FT across various metrics, including NSE, KGE, Pearson-r, FHV, FLV, and PBias.

Ratio (%)	NSE	KGE	Pearson-r	FHV	FLV	PBias
improved	68.7	54.4	75.0	49.5	51.4	49.7
decreased	31.3	45.6	25.0	50.5	48.6	50.3

References

- Addor, N., Newman, A.J., Mizukami, N. and Clark, M.P. 2017. The CAMELS data set: catchment attributes and meteorology for large-sample studies. *Hydrology and Earth System Sciences* 21(10), 5293-5313.
- Frame, J.M., Kratzert, F., Klotz, D., Gauch, M., Shalev, G., Gilon, O., Qualls, L.M., Gupta, H.V. and Nearing, G.S. 2022. Deep learning rainfall–runoff predictions of extreme events. *Hydrology and Earth System Sciences* 26(13), 3377-3392.
- Gupta, H.V., Kling, H., Yilmaz, K.K. and Martinez, G.F. 2009. Decomposition of the mean squared error and NSE performance criteria: Implications for improving hydrological modelling. *Journal of hydrology* 377(1-2), 80-91.
- Kratzert, F., Klotz, D., Herrnegger, M., Sampson, A.K., Hochreiter, S. and Nearing, G.S. 2019a. Toward improved predictions in ungauged basins: Exploiting the power of machine learning. *Water Resources Research* 55(12), 11344-11354.
- Kratzert, F., Klotz, D., Shalev, G., Klambauer, G., Hochreiter, S. and Nearing, G. 2019b. Towards learning universal, regional, and local hydrological behaviors via machine learning applied to large-sample datasets. *Hydrology and Earth System Sciences* 23(12), 5089-5110.
- Liu, J., Bian, Y., Lawson, K. and Shen, C. 2024. Probing the limit of hydrologic predictability with the Transformer network. *Journal of Hydrology* 637, 131389.
- Nash, J.E. and Sutcliffe, J.V. 1970. River flow forecasting through conceptual models part I—A discussion of principles. *Journal of hydrology* 10(3), 282-290.
- Yilmaz, K.K., Gupta, H.V. and Wagener, T. 2008. A process-based diagnostic approach to model evaluation: Application to the NWS distributed hydrologic model. *Water resources research* 44(9).